

1. Project Title

Ecommerce & Point of Sales Solution

2. Group Members

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3. Introduction / Overview

This project is about creating a comprehensive system that allows a business to sell products both online (through an e-commerce website) and in a physical shop (using a Point of Sales, or POS, system). We are integrating both sales channels so that all products, inventory levels, and sales data are managed from one central place. The primary goal is to make business management easier and keep track of stock accurately and efficiently.

4. Scope of the Project

The project scope covers the design, development, and testing of the end-to-end system, including:

- **E-commerce Frontend:** A publicly accessible website for product browsing, shopping, and online order placement.
- **POS Backend/Interface:** A dedicated application or interface for managing physical store transactions, including barcode scanning and cash/card payments.
- **Centralized Management System:** A secure admin dashboard for inventory control, order fulfillment, user management, and detailed sales reporting.
- **Integration:** Seamless, real-time synchronization between online orders, POS sales, and centralized inventory.

5. System Features / Functional Requirements

- **Product Management:** CRUD operations for products, including categorization, pricing, and variant support.
- **Inventory Control:** Real-time stock level tracking, low-stock alerts, and stock reconciliation features.
- **Online Ordering:** Customer registration, shopping cart functionality, secure payment gateway integration, and order tracking.
- **POS Transactions:** Quick item lookup, barcode scanning support, tax calculation, split payments, and end-of-day closing reports.
- **User/Role Management:** Defining different access levels for administrators, store managers, and sales associates.
- **Reporting:** Generating sales reports, inventory movement reports, and profit/loss statements.

6. Pages / Screens (Frontend Design)

- **E-commerce Site:** Home Page, Product Listing Page, Product Detail Page, Shopping Cart, Checkout Page, Customer Profile/Order History.
- **POS Interface:** Login Screen, Main Sales/Transaction Screen, Product Lookup, Hold/Retrieve Transaction Screen, End-of-Day Report.
- **Admin Dashboard:** Dashboard Overview, Product Management, Inventory Management, Order Management, User Management, Reports.

7. Database Design (Only Table Names)

- Users
- Products
- Product_Categories
- Inventory
- Online_Orders
- POS_Transactions
- Order_Items
- Customers
- Payments

8. Tools & Technologies

- **Frontend:** HTML, CSS, and JavaScript
- **Backend:** [Node.js](#) (For Online Hosting we can use AWS free tier or Render)
- **Database:** PostgreSQL (locally) or Supabase/Neon.tech/Aiven (for online deployment)
- **Deployment:** Local or Vercel/Netlify
- **Version Control:** GitHub